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Conference 2000


Next Generation of Palm Expansion Technologies:
Get Plugged In at Palm
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# Why Are We Here?

- Market is demanding expansion solutions
- Create expansion products for the world's leading handheld computers
- Capitalize on this market opportunity NOW!!!
- And…




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# Mission Statement

- Create a world class expansion architecture
- Grow the Palm Economy by providing expansion opportunities
- Enable developers by creating comprehensive support program for all phases
- Create marketing and distribution vehicles for your products

# Agenda

- Dual Expansion Architecture
- Technology Demonstrations
- PluggedIn Program
- Q&A

# Why a Dual Expansion Architecture

- Allows for wearable devices with expansion
- Based on a robust industry standard
- Interoperable with other consumer devices
- I/O capable
- Fast transfer rate 10MB/sec
- USB capable











# Comparison of Media Types

| | SDCard | Memory Stick | Compact Flash | Springboard |
|---|---|---|---|---|
| Open Standard | YES (Supported by 153 Comapnies) | No | Yes | No |
| Data Transfer Speed (Max.)* | 10MB/s | 2.45 MB/s | 2.2 MB/s | 11 MB/s |
| SPI Support | Yes | No | No | No |
| SDMI Security | Version 2.0 | Version 2.0 [1,2] | Planned | None |
| Volume/Size (cubic mm) | 1,612mm | 3,010mm | 5,141mm | 15,325mm |
| Write Protect | Physical | Physical | Electrical | None |




Notes:
1. MagicGate on select higher capacities only
2. Not aligned with DVD Encryption Standard
*Also “Duo” version without I/O support.
*Theoretical maximum, rates may vary under certain conditions
SPI-Serial peripheral interface
Compact media illustrations not to scale or actual size

# Dual Expansion Architecture

- Expansion card slot
  - Why we chose SD and MMC:
    - Size (Think: postage stamp)
    - I/O (for Bluetooth, Ethernet, modems, cameras, etc.)
    - Security (supports digital rights management)
    - Flexibility (used in cameras, video recorders, MP3's, etc.)
    - Open platform (no licensing fees = lots of third-party solutions)
    - Cost (less expensive media because of volume)





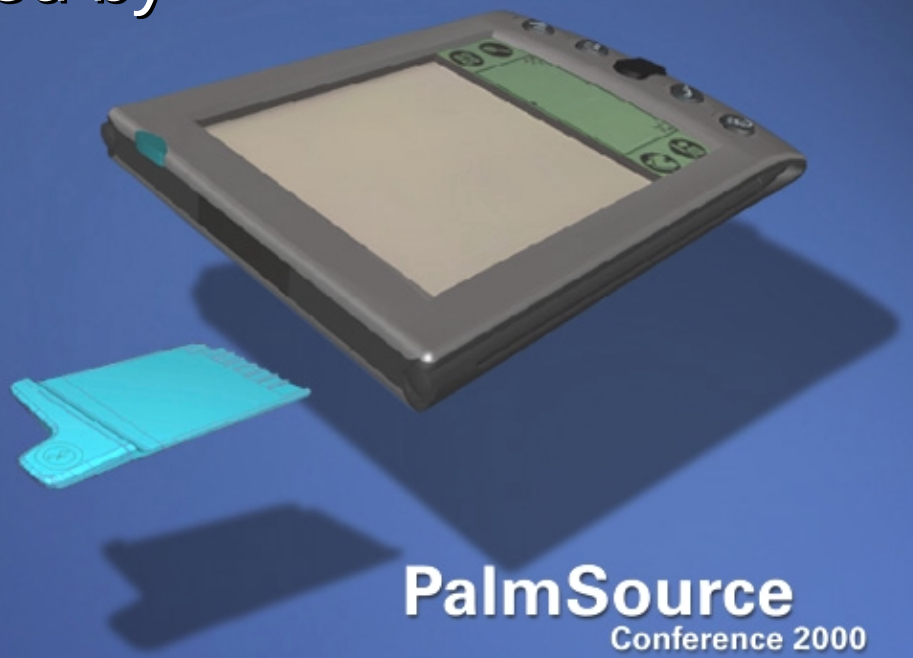

# Dual Expansion Architecture

- Universal Connector
  - Create one product for Palm's future product family
  - Enables plug & play convenience
  - Can work in concert with SD or MMC cards
  - USB capable



# Expansion Card Slot

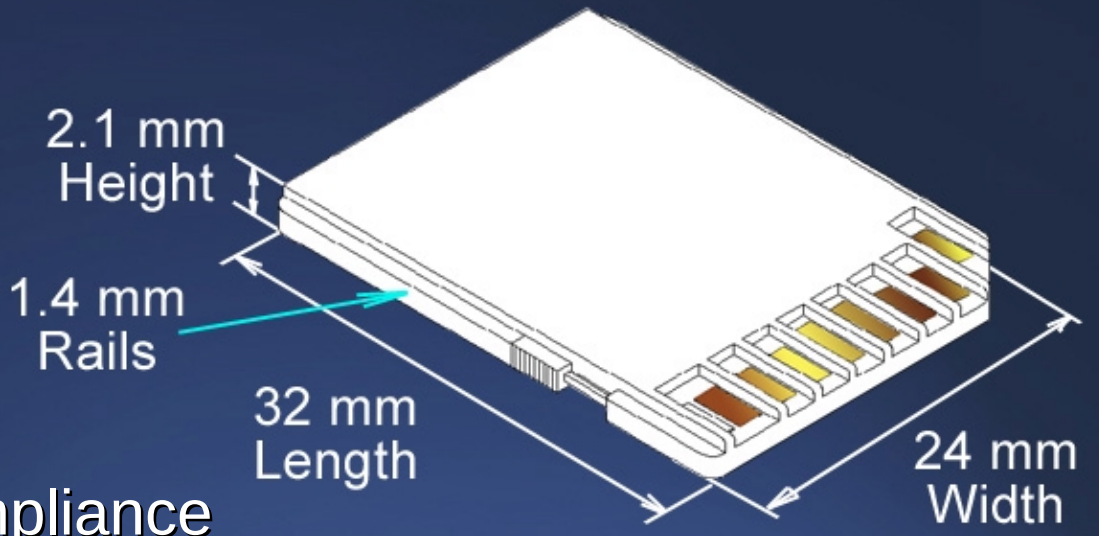
- Supports SD, MMC and SDIO
- Interoperable between SD enabled products
- SDIO combination card: IO function + memory
- Industry standard backed by over 150 companies



# SD Interoperability



# SD Cards




- Small form factor
- Open Standard
- SDMI version 2.0 compliance
- 10MB/sec data transfer rate (max)
- Write-protect tab on card casing
- High storage capacity (est. >1GB by 2002)
- Forward Compatibility between MultiMedia Card and future SD hosts
- Robust
  - High electrostatic-discharge tolerance
- Ability to add miniature I/O devices
  - Bluetooth

# SD/MMC Comparison

## Major Differences

| | SD | MultiMediaCard |
|---|---|---|
| Capacities | >32MB (2x MMC) | 4-64MB |
| Bus Xfer Speed (Max.) | 10MB/s | 2.5MB/s |
| Height | 2.1mm | 1.4mm |
| SDMI Security | Version 2.0 | Version 1.0 |
| Operating Voltage Roadmap | 3.6 to 2.7V | 3.6 to 2.7V[1] |
| Volume/Size | 1,612mm | 1,075 mm |
| Weight | 2.0 grams | 1.5 grams |
| ESD[2] | +/- 10KV | +/- 4KV |




Notes: 1 1.3V and 1.8V Under Active Discussion in the MMCA
2 Discharge at I/F with Ball Probe

# SD/MMC Co-Existence

## Shared Marketplace



- MMC = Primarily "Non-secure" applications:
  - Digital Still and Video Cameras, HPCs
  - Many Cellular, GPS, E-book applications
  - SDMI 1.0 Compliant Audio Players
  - ROM versions for low-cost distribution

+

- SD = Primarily "Secure" Audio/Video applications
  - SDMI 2.0 Compliant Audio Players
  - Additional Cellular, GPS and E-book apps
  - Future Personal Video Players—2002






# SD Demonstration

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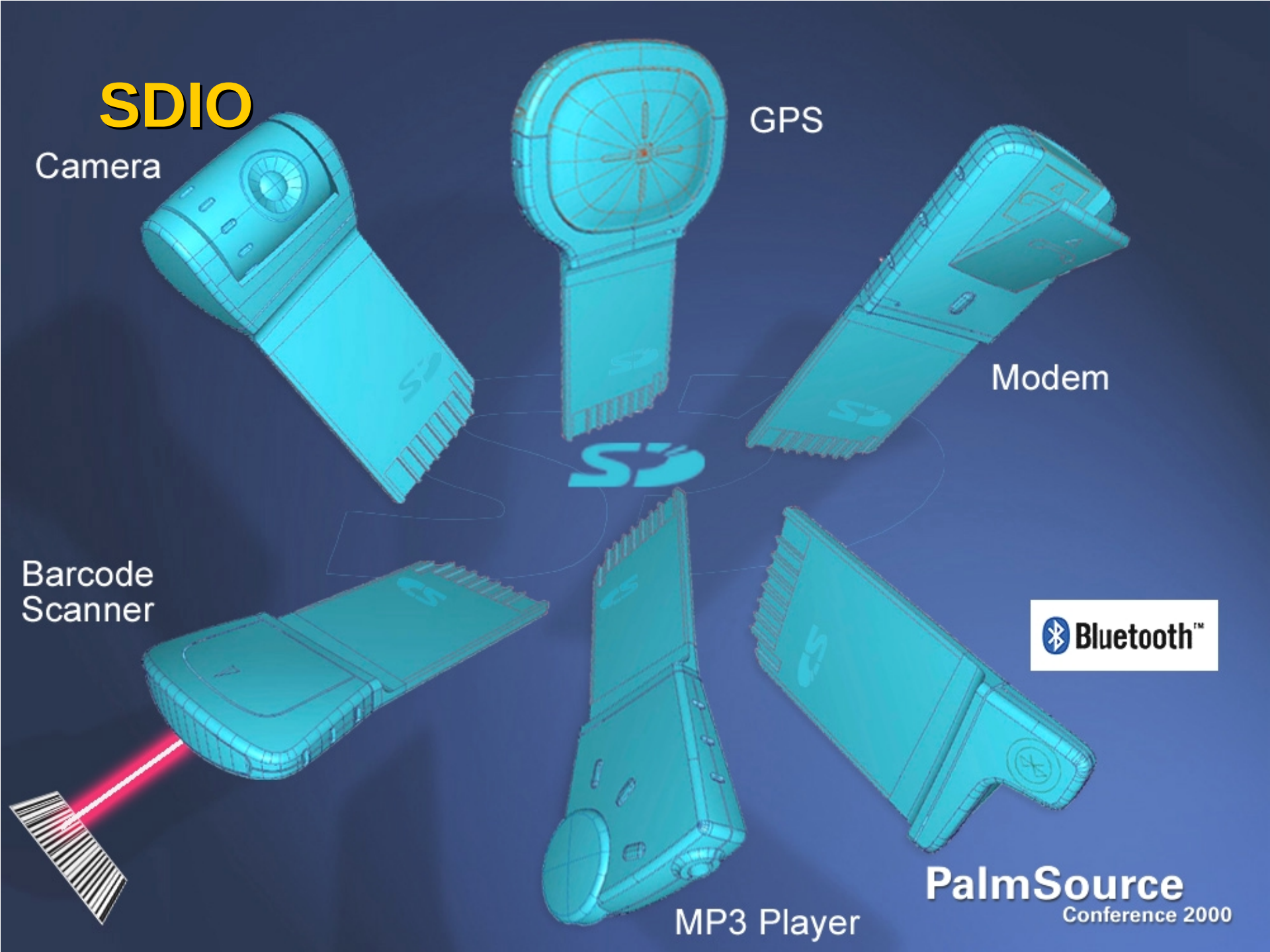

# SDIO

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# SDIO

- Expandable SD Card
  - Alleviates slot contention for memory and I/O users
  - I/O only possible for cost-sensitive users
  - Full adherence to SDA specifications for I/O and combination card functionality
  - Maximum flexibility for users with multiple I/O needs




SDIO
Camera
GPS
Modem
Barcode
Scanner
Bluetooth™
MP3 Player
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# SDIO Demonstration

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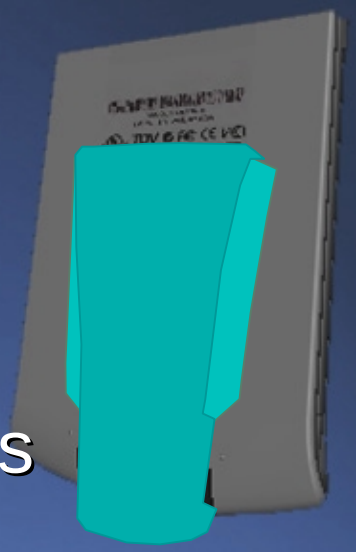



# Universal Connector

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# Universal Connector Expansion



- Common electrical and mechanical interface for peripherals
- Enables broad range of accessories over a large portfolio of Palm products
- Enables plug-and-play convenience
- USB connectivity for faster transfers
- RS232 connectivity to simplify interfacing
- Serial rates up to 230 Kbaud
- EDK ensures ease of development




# Universal Connector PnP Demonstration

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# PluggedIn Program

## The Complete Developer Resource For Palm™ Expansion Solutions

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# Enabling the Palm Economy

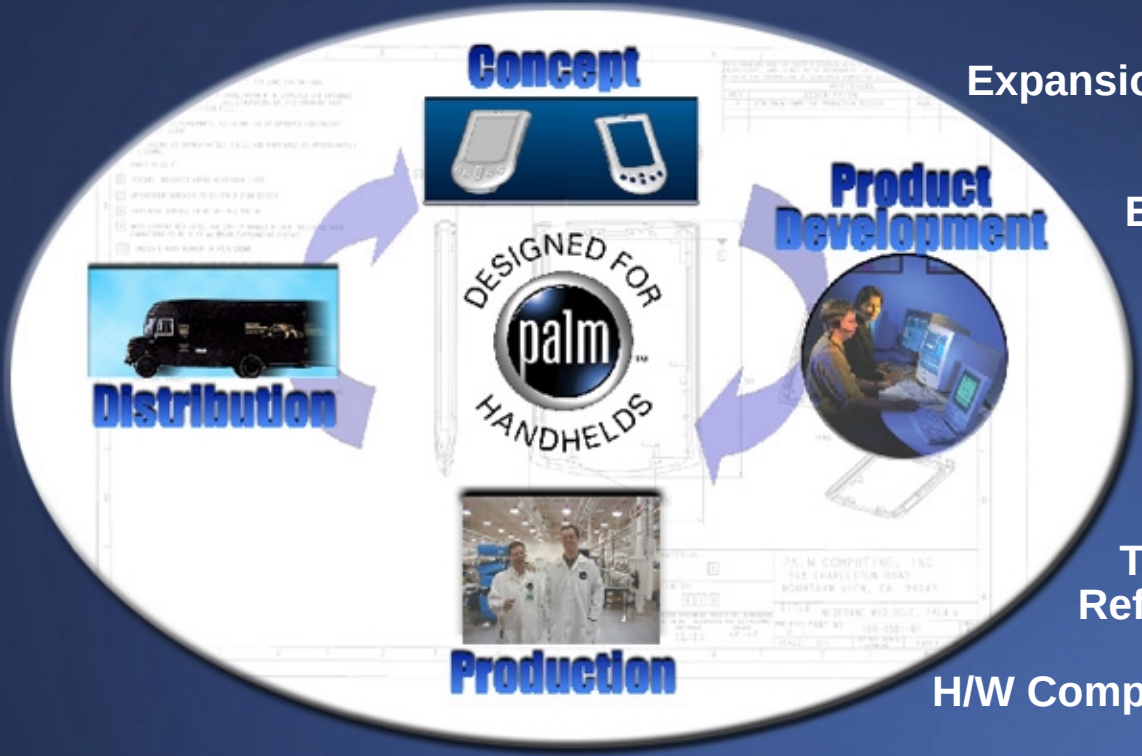
Evangelize Palm's expansion technologies
Advanced Product Notification
Developer Tools

Select 3rd party products on Palm.com

Access to SD/MMC republishers

Solutions Marketplace Access

Marketing Opportunities




Expansion Development Kits

Expansion Store

Early units for testing

Technical Support

Tooling Services
Reference Designs

H/W Compatibility Testing

Assistance from premiere manufacturing partners
Relationships for SD/MMC manufacturing
ROM Flashing Services



Distribution



Production



Product Development



Concept




1. Log onto www.palmos.com

2. Get Plugged In at:
pluggedin@palm.com

PluggedIn Program

The Complete Developer Resource For Palm® Expansion Solutions



3. Be introduced to the Business Development Team

DESIGNED FOR palm HANDHELDS

4. Become Designed for Palm

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Q&A




# Thank You

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